

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Ch.Poorna Teja	Reg. No.:	192472149
Section / Batch:	2023	Date:	28/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Q47. A communication network calculates distances between nodes to identify the closest pair using brute-force computation.

- Explain how all node pairs are evaluated.**
- Derive the time complexity.**
- Discuss scalability issues.**

Code of Conduct

I ch.poorna teja(192472149) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ch.poorna teja

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q47. A Communication Network Calculates Distances Between Nodes to Identify the Closest Pair Using Brute-Force Computation

Problem Interpretation

In a communication network, each node represents a device or communication point, and the distance between two nodes represents the communication cost or physical separation. The objective is to identify the pair of nodes with the minimum distance so that communication can be optimized. The Brute Force Closest Pair algorithm solves this problem by calculating the distance between every possible pair of nodes and selecting the pair with the smallest distance. Although simple and guaranteed to produce the correct result, this approach becomes computationally expensive as the number of nodes increases.

a) How All Node Pairs Are Evaluated

The brute-force algorithm compares every possible pair of nodes in the network.

Step-by-step process:

1. Consider the first node and compute its distance to every other node.
2. Store the smallest distance found so far.
3. Move to the second node and compare it with all remaining nodes.
4. Continue this process until every unique pair has been evaluated.
5. After all comparisons, return the pair with the minimum distance.

If there are n nodes, the total number of unique pairs is:

$$\text{Number of pairs} = n(n - 1) / 2$$

Example:

Suppose the network has six nodes:

A, B, C, D, E, F

The brute-force algorithm evaluates the following pairs:

A-B
A-C
A-D
A-E
A-F
B-C
B-D
B-E
B-F
C-D
C-E
C-F
D-E
D-F
E-F

Total pairs evaluated = $6 \times 5 / 2 = 15$ pairs.

For each pair, the Euclidean distance is calculated using:

$$\text{Distance} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

The algorithm compares each computed distance with the current minimum distance. Whenever a smaller distance is found, it updates the closest pair. After evaluating all pairs, the minimum distance obtained is the final answer.

Because every possible pair is examined exactly once, the brute-force algorithm always produces the correct closest pair.

b) Time Complexity Analysis

Case	Description	Complexity
Best Case	Every pair must still be examined	$O(n^2)$
Average Case	Distance computed for every possible pair	$O(n^2)$
Worst Case	Every possible pair compared	$O(n^2)$

The total number of comparisons is:

$$n(n - 1) / 2$$

For example:

- $n = 100$ nodes $\rightarrow 4,950$ comparisons
- $n = 1,000$ nodes $\rightarrow 499,500$ comparisons
- $n = 10,000$ nodes $\rightarrow 49,995,000$ comparisons

Since the number of comparisons increases proportionally to n^2 , the time complexity of the brute-force closest pair algorithm is:

$$\text{Time Complexity} = O(n^2)$$

The algorithm requires only a few variables to store the current minimum distance and the closest pair.

$$\text{Space Complexity} = O(1)$$

c) Scalability Issues

Although the brute-force approach is simple and guarantees the correct result, it does not scale well for large communication networks.

Major scalability issues include:

1. Rapid Growth in Comparisons

The number of distance calculations increases quadratically with the number of nodes. Doubling the number of nodes approximately quadruples the amount of work.

2. Increased Execution Time

For small communication networks, brute force performs efficiently. However, for networks containing thousands or millions of nodes, computing every pairwise distance becomes extremely time-consuming.

3. High Computational Cost

Every distance calculation involves arithmetic operations such as subtraction, multiplication, addition, and square root computation. Performing millions of such calculations significantly increases processing time.

4. Unsuitable for Real-Time Systems

Modern communication networks frequently add, remove, or move nodes. Recomputing every possible pair after each update is impractical for real-time routing or monitoring systems.

5. Better Alternatives for Large Networks

For large datasets, more efficient algorithms such as the Divide and Conquer Closest Pair Algorithm solve the same problem in $O(n \log n)$ time, making them much more suitable for practical applications.

Justification and Recommended Approach

The choice of algorithm depends on the size of the communication network.

Scenario	Recommended Approach	Reasoning
Small communication networks	Brute Force Closest Pair	Simple, easy to implement, guaranteed correct
Medium-sized networks	Divide and Conquer Algorithm	Faster than brute force
Large communication networks	$O(n \log n)$ Closest Pair Algorithm	Efficient and scalable for large datasets

Conclusion

The Brute Force Closest Pair algorithm evaluates every possible pair of nodes by computing the distance between them and selecting the minimum. For n nodes, it performs $n(n - 1)/2$ comparisons, resulting in $O(n^2)$ time complexity and $O(1)$ space complexity. While the method is simple, exhaustive, and always correct, it becomes inefficient for large communication networks because the number of comparisons grows rapidly. Therefore, brute force is best suited for small networks, whereas Divide and Conquer algorithms with $O(n \log n)$ complexity are preferred for large-scale communication systems due to their superior scalability and efficiency.